DEMENTIA PREDICTION USING LOGISTIC REGRESSION

A Research Report

Submitted by:
Kabita Joshi
Roll No: 12

Submitted to:
Pravat Uprety

School of Mathematical Sciences
Tribhuvan University
Kathmandu, Nepal

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1 Introduction

Dementia is a term used to describe a group of symptoms that affect memory, thinking, and social abilities severely enough to interfere with daily functioning. It is not a disease in itself but rather the result of various diseases and conditions that damage the brain. The most common form of dementia is Alzheimer's disease, followed by vascular dementia, Lewy body dementia, and frontotemporal dementia. Each type has its own set of symptoms and progression patterns, but all involve a decline in cognitive abilities beyond what might be expected from normal aging.

The causes of dementia are varied and can include age-related brain changes, stroke, head injuries, and genetic factors. While dementia mostly affects older adults, it is not a normal part of aging. Risk factors such as high blood pressure, diabetes, smoking, and an inactive lifestyle can increase the likelihood of developing dementia. Early symptoms often include memory loss, confusion, difficulty finding words, and changes in behavior. As the condition progresses, individuals may require help with basic tasks and constant care.

Although there is currently no cure for dementia, treatments and lifestyle adjustments can help manage symptoms and improve quality of life. Medications may be prescribed to slow cognitive decline or manage behavioral changes, and support from healthcare providers, caregivers, and community resources is essential. Increasing awareness and understanding of dementia can reduce stigma and ensure that people living with the condition receive the support and respect they need.

The Mini-Mental State Examination (MMSE) is a widely used screening tool for cognitive impairment, particularly in the context of dementia or other neurological conditions. It's a brief questionnaire that assesses various aspects of cognitive function, including orientation, memory, attention, and language. While often used to screen for dementia, the MMSE is not a diagnostic test and should be considered alongside other clinical evaluations

2 Problem Statement

Early and accurate detection of dementia is critical for timely intervention, treatment planning, and improving the quality of life for patients and their caregivers. However, dementia is often diagnosed in its later stages due to the subtle and gradual onset of symptoms, making early detection challenging. Traditional diagnostic methods, which rely heavily on clinical assessments, cognitive tests, and neuroimaging, can be time-consuming, expensive, and not always accessible, especially in low-resource settings. Key research questions of this study are:

- Why is dementia often diagnosed in its later stages, and how can early detection be improved using data-driven approaches?
- Can we develop a reliable and accurate predictive model that uses clinical, cognitive, or

demographic data to identify individuals at risk of developing dementia?

- How can such a predictive tool support healthcare professionals in early intervention and improve outcomes for patients and caregivers?

3 Objectives of the Study

The major objective of the study is to predict dementia using . The specific objectives of the study are as follows: follows:

- To identify key features and risk factors from clinical, cognitive, and demographic data that contribute to the early onset of dementia.
- To develop and validate a predictive model using machine learning or deep learning techniques for early and accurate detection of dementia.
- To create a user-friendly, interpretable tool that supports healthcare professionals in decision-making, enabling timely interventions and improving patient outcomes.

4 Hypothesis of the Study

The research hypothesis of this study is given below:
Null Hypothesis (H_0): There is no significant impact of Age, Education, Mobility, Body Height, Mini Nutritional Assessment Score, Body Weight, Waist Size, BMI and Hyperlipidaemia status on Dementia (MMSE Class).
Alternative Hypothesis (H_1): There is a significant impact of Age, Education, Mobility, Body Height, Mini Nutritional Assessment Score, Body Weight, Waist Size, BMI and Hyperlipidaemia status on Dementia (MMSE Class).

5 Rationale of the Study

Dementia is a growing public health concern globally, with rising prevalence due to aging populations. Early detection is crucial to managing the disease effectively, slowing its progression, and improving patient quality of life. However, current diagnostic methods are often complex, expensive, and applied too late in the disease course. This creates a critical need for more accessible and timely predictive tools.

The rationale behind this study is to explore how clinical, demographic, nutritional, and physical variables—such as age, education, BMI, mobility, and Mini Nutritional Assessment (MNA) scores—can be used to develop a predictive model for dementia. These variables are relatively easy to collect and may serve as important early indicators of cognitive decline. By

identifying patterns in these data, the study aims to enable earlier intervention and reduce the burden of undiagnosed or late-diagnosed dementia.

Ultimately, this research seeks to contribute to the development of a cost-effective, interpretable, and practical screening tool that can be integrated into routine clinical assessments. Such a tool would support healthcare professionals in identifying at-risk individuals, especially in resource-limited settings, thereby facilitating proactive and personalized care. .

6 Brief Review of Literature

Early detection of dementia is crucial for effective intervention and management. Traditional diagnostic methods often identify the condition at advanced stages, limiting treatment options. Recent research has focused on leveraging machine learning (ML) and artificial intelligence (AI) to predict dementia earlier and more accurately.

Li et al. (2019) developed a deep learning model using hippocampal MRI data to predict the progression from mild cognitive impairment (MCI) to Alzheimer's disease (AD). Their model achieved a concordance index of 0.762, indicating strong predictive capability. Similarly, Jo et al. (2019) reviewed various deep learning approaches for AD diagnosis, highlighting that combining multimodal neuroimaging data can enhance classification accuracy.

Tan et al. (2023) demonstrated that integrating socio-demographic, vascular risk factors, and neuroimaging markers using ML classifiers like support vector machines (SVM) and gradient boosting machines (GBM) can effectively diagnose cognitive impairment. Their ensemble model achieved an accuracy of 83%, outperforming individual classifiers. Mamidala (2023) proposed a holistic dementia risk prediction model that combines ML with network systems biology, achieving a sensitivity of 85% and specificity of 99%.

Nutritional status has also been linked to dementia risk. A study published in Clinical Nutrition found that low Mini Nutritional Assessment-Short Form (MNA-SF) scores were associated with higher mortality rates among older adults with dementia, emphasizing the importance of nutritional assessments in predicting dementia outcomes. Morris et al. (2015) introduced the MIND diet, which combines elements of the Mediterranean and DASH diets, and found it to be associated with a reduced incidence of AD.

Innovative tools are being developed for early dementia detection. Researchers at Queen Mary University of London created a test using functional MRI to predict dementia up to nine years before diagnosis with over 80% accuracy. Additionally, the Florey Dementia Index, developed by Australian researchers, uses online surveys and age analysis to predict the onset age of early Alzheimer's disease.

These studies collectively underscore the potential of integrating ML techniques with clinical, demographic, and lifestyle data to enhance early dementia prediction. Such approaches can facilitate timely interventions, improving patient outcomes and reducing the burden on healthcare systems.

7 Research Methodology

A research methodology is an outline of how a given piece of research is carried out. It includes the research design, the target population, sampling method and sample size, data collection procedures and data analysis method.

7.1 Research Design

The framework, structure, and strategy of an inquiry created to find the answer to a research question and to manage deviations is known as a research design. This study aims to analyse the factors affecting dementia status. This study uses quantitative research design with both descriptive and inferential statistics.

7.2 Population and Sample

The dataset used for this study is the Dementia Dataset. The sampling process to be used for this study is random sampling.

7.3 Nature and Sources of Data

There are three sources of data. They are primary, secondary, and tertiary data. Primary sources of data come from field survey, questionnaires, interviews etc. Secondary sources of data come from published information, books, reports, journals etc. This study relies on the analysis of secondary data that was recorded to store the status of MMSE among various test subjects.

7.4 Research Framework and Definition of Variables

Conceptual framework helps to clearly identify the variables that are used in the research process and shows how particular variables are connected with each other in the study. The conceptual framework along with the depiction of the variables used in this study are given below:

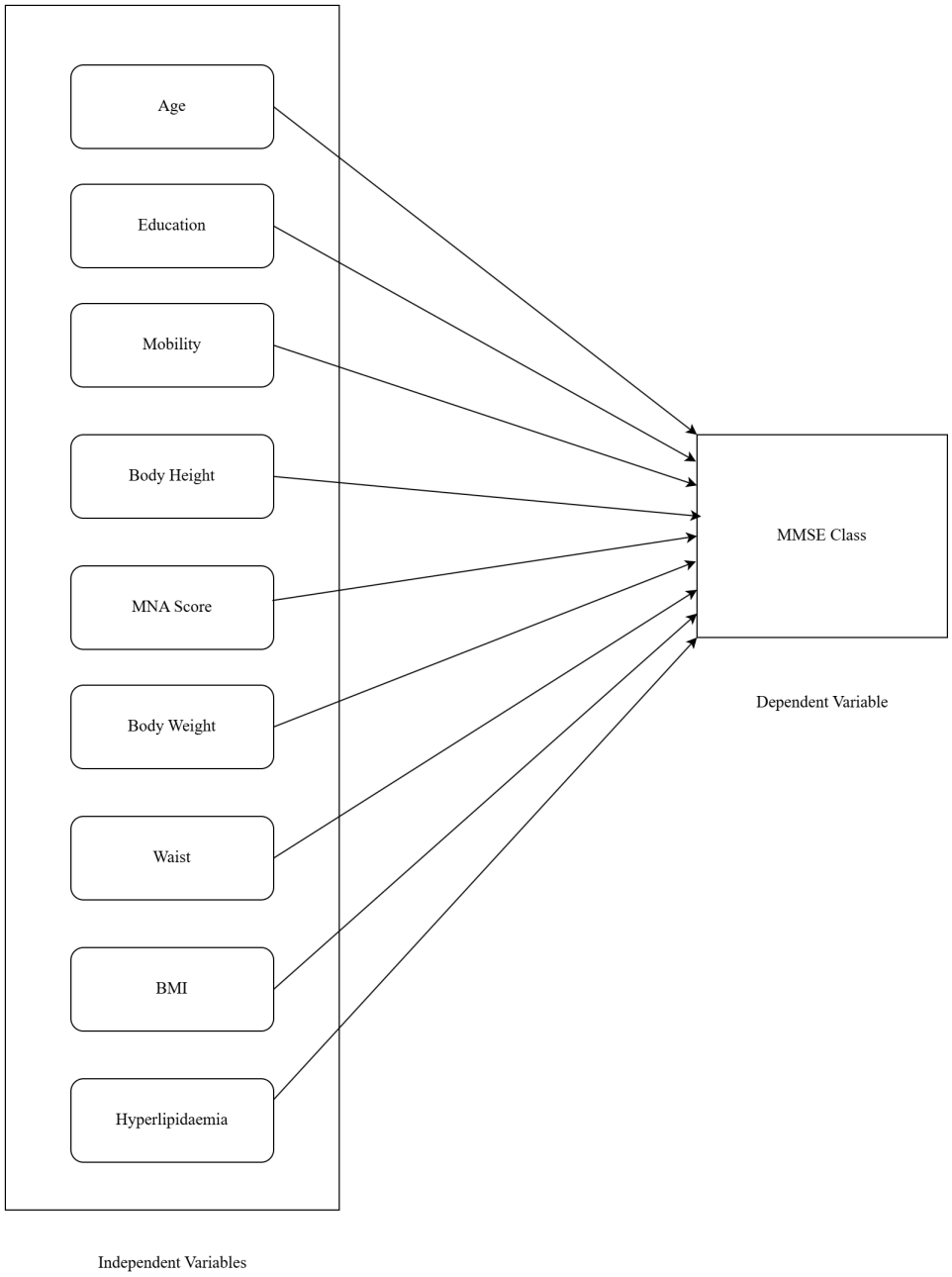

Figure 1: Conceptual Framework: Dementia Prediction

7.5 Research Model

The logistic regression model employed in this study tries to analyze the impact of impact of Age, Education, Mobility, Body Height, Mini Nutritional Assessment Score, Body Weight,

Waist Size, BMI and Hyperlipidaemia status on Dementia (MMSE Class). From the conceptual framework, the model take the following form: The functional form of the MMSE Class is given by:

$$\begin{aligned} MMSE = f(&Age, Education, Mobility, \\ &Body\ Height, MNA\ Score, Body\ Weight, \\ &Waist\ Size, BMI, Hyperlipidaemia) \end{aligned} \tag{1}$$

More specifically, logistic model can be expressed as:

$$\begin{aligned} \ln \frac{P_i}{1-P_i} = &\beta_0 + \beta_1 A + \beta_2 E + \beta_3 M + \beta_4 BH + \beta_5 MNAaq3 \\ &+ \beta_6 BW + \beta_7 MNAbTot + \beta_8 W + \beta_9 BMI + \beta_{10} MNAaTot + \beta_{11} Hyper \end{aligned} \tag{2}$$

Where:

- P_i = Probability of Getting Dementia (MMSE = 1)
- A = Age
- BH = Body Height
- $MNAaq3$ = Score on MNA test, part A, Q3
- BW = Body Weight
- $MNAbTot$ = Score on MNA test, Part B, Total
- W = Waist Size
- BMI = Body Mass Index
- $MNAaTot$ = Score on MNA test, Part A, Total
- $Hyper$ = Hyperlipidaemia Status
- β_0 is the constant term
- β are the coefficients of regression

7.6 Methods of Analysis

To assess the data, several financial, mathematical, and statistical tools are employed. For testing the significant impact of independent variables on dependent variable, Likelihood test is performed using LR statistic. In order to test for the impact of individual variable on the dependent variable, Wald's Test has been used.

8 Results and Discussion

Data presentation and analysis are fundamental components of academic research, as well as various commercial, industrial, and marketing endeavors, along with professional practices. Analyzing data aids in understanding its implications and facilitates decision-making or addressing research inquiries. This phase of the study involves the presentation and examination of gathered data, often utilizing tools such as tables and visual aids like bar diagrams or figures.

8.1 Data Presentation

The core objective of this research is to analyze the impact of various medical factors on Dementia (MMSE).

ID	Age	Edu_ID	Mobility	Height	MNAa_q3	Weight	MNAb_tot	Waist	BMI	MNAa_tot	Hyper	MMSE_class
0	86	2	4	147.0	2	39.1	8.8	73.8	18.2	10	NA	0
1	69	2	4	169.0	2	60.4	13.4	86.9	25.5	14	0	0
2	65	3	4	165.0	2	76.7	13.5	90.5	26.7	14	0	0
3	65	2	4	173.0	2	60.4	12.5	96.0	26.6	NA	0	0
4	77	2	4	143.0	2	58.0	12.0	97.9	28.7	14	0	0
5	63	3	4	171.0	2	79.6	13.4	94.8	24.4	14	0	0

Table 1: Sample Data from Dementia Dataset

The dataset contains 13 variables across 6 individuals, capturing demographic, physical, and nutritional attributes relevant to dementia prediction. Variables include age, education level, mobility, body measurements (height, weight, BMI, waist), and Mini Nutritional Assessment (MNA) scores. The target variable, ‘MMSE_class‘, indicates dementia classification, while ‘Hyperlipidaemia‘ is a binary health indicator. Some entries contain missing values (‘NA‘), which may require preprocessing before analysis.

After removing the NA and NULL values from the dataset, the summary statistics was run and the results are tabulated as follows:

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
ID	1	720.5	1466.0	1447.2	2160.5	2872.0
Age	57.0	68.0	79.0	77.82	87.0	104.0
Education_ID	-	-	-	-	-	-
Mobility	-	-	-	-	-	-
body_height	132.1	146.1	151.2	153.3	160.8	184.0
MNAa_q3	0.0	2.0	2.0	1.991	2.0	2.0
body_weight	30.2	47.4	55.06	56.24	64.64	97.41
MNAb_tot	6.0	10.55	12.5	11.9	13.0	15.0
waist	57.5	77.8	85.6	85.28	93.0	118.0
BMI	13.44	20.32	23.3	23.51	26.34	40.66
MNAa_tot	7.0	12.0	14.0	13.06	14.0	14.0
Hyperlipidaemia	0 (n=1772)	-	-	-	-	1 (n=91)
MMSE_class	0 (n=1542)	-	-	-	-	1 (n=321)

Table 2: Summary Statistics of Cleaned Dementia Dataset

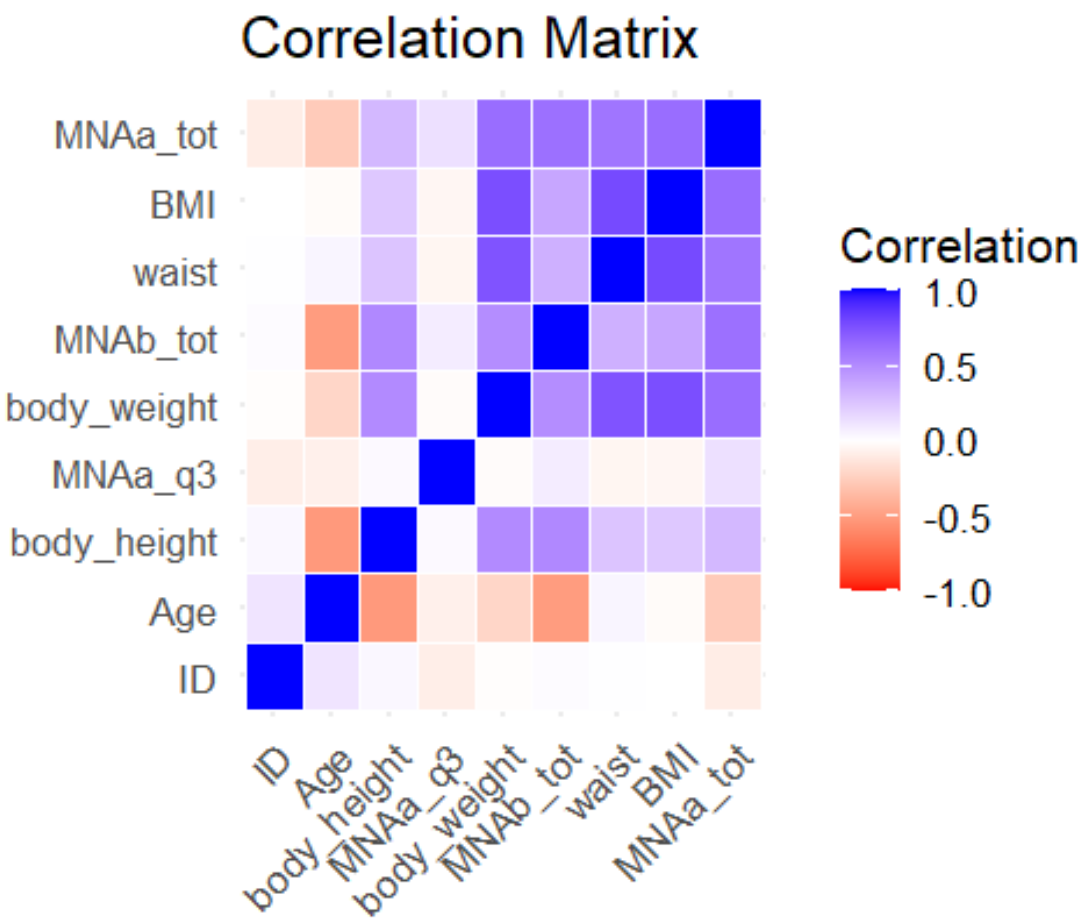

Figure 2: Correlation matrix of numerical features in the dementia dataset

The diagram above represents the correlation among the numerical features in the dementia

dataset. The correlation is represented in terms of a heatmap.

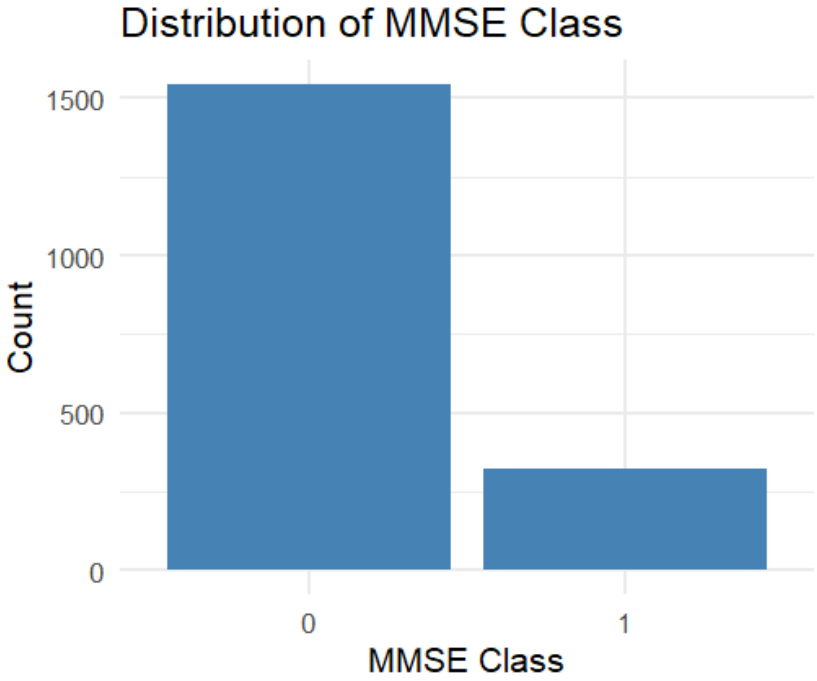

Figure 3: Distribution of the Target Class

The diagram above represents the distribution of the target class (MMSE). The value 1 represents the presence of Dementia whereas 0 represents the absence.

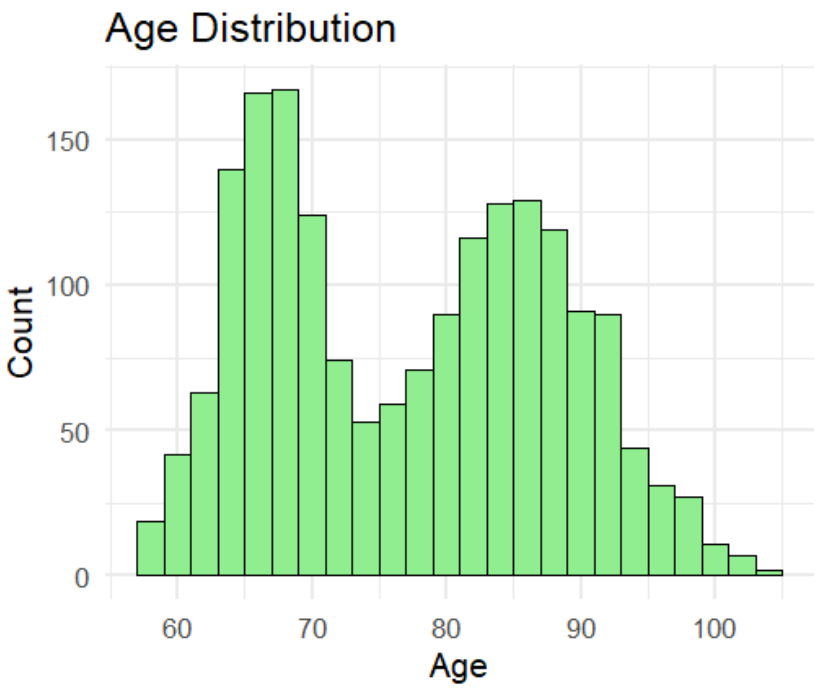

Figure 4: Distribution of the Age Variable

The diagram above represents the distribution of age variable. It doesn't exhibit the prop-

erties of normal distribution. It shows two peaks.

8.2 Regression Model and Statistical Tests

Logistic regression is a statistical model used to predict the probability of a binary outcome (such as 0 or 1) based on one or more predictor variables. It uses the logistic function to model the relationship between the dependent variable and independent variables. Unlike linear regression, it outputs probabilities that are then mapped to binary classes. The results of running logistic regression for the given dataset are shown in table below:

Model	**Residual DF**	**Residual Deviance**	**DF**	**Deviance**	**Pr(>Chi)**
Model 1: Null Model	1304	1199.80	–	–	–
Model 2: Full Model	1289	531.68	15	668.12	$< 2.2 \times 10^{-16}$

Table 3: Likelihood Ratio Test comparing null model and full logistic regression model

The Likelihood Ratio Test (LRT) compares two logistic regression models to assess whether the inclusion of multiple predictors significantly improves model performance. In this case, Model 1 is a null model containing only the intercept, with a residual deviance of 1199.80 and 1304 residual degrees of freedom. Model 2 is a full model that includes predictors such as ID, Age, Education_ID, Mobility, body height, MNAa_q3, body weight, MNAb_tot, waist, BMI, MNAa_tot, and Hyperlipidaemia. This model has a residual deviance of 531.68 with 1289 degrees of freedom, and the difference in deviance between the two models is 668.12 with 15 degrees of freedom.

The associated p-value for the deviance reduction is less than 0.05, which is highly statistically significant. This means that the full model offers a much better fit than the null model, indicating that the included predictors collectively contribute meaningful information for predicting the outcome variable, ‘MMSE_class‘, which classifies dementia status. Therefore, we can conclude that demographic, physical, and nutritional factors have a significant association with the risk of dementia in this dataset.

The results of individual test of predictor variables on the dependent variable is given below:

Variable	Estimate	Std. Error	z value	Pr(>|z|)	Significance
(Intercept)	17.78	1497.00	0.012	0.99053	
ID	0.000259	0.000142	1.824	0.06808	.
Age	0.08651	0.01803	4.797	0.00000161	***
Education_ID2	-3.373	0.3676	-9.178	<2e-16	***
Education_ID3	-2.357	0.9222	-2.556	0.01060	*
Education_ID4	-15.01	724.9	-0.021	0.98348	
Mobility3	-12.01	1497.00	-0.008	0.99360	
Mobility4	-11.68	1497.00	-0.008	0.99378	
body_height	-0.03844	0.02205	-1.743	0.08131	.
MNAa_q3	-1.941	0.8156	-2.380	0.01732	*
body_weight	-0.02358	0.02155	-1.094	0.27398	
MNAb_tot	-0.6161	0.1022	-6.030	0.00000000164	***
waist	0.04908	0.01816	2.703	0.00688	**
BMI	0.1003	0.05020	1.998	0.04573	*
MNAa_tot	-0.2275	0.1180	-1.929	0.05376	.
Hyperlipidaemia1	-0.5200	0.5802	-0.896	0.37012	

Signif. codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$

Table 4: Wald test results for logistic regression coefficients predicting dementia status (MMSE_class).

The logistic regression analysis reveals that several predictors significantly influence the likelihood of dementia status (MMSE_class). Age shows a strong positive association ($p < 0.001$), indicating that higher age increases dementia risk. Education levels (specifically Education_ID2 and Education_ID3) have significant negative coefficients, suggesting that higher education is associated with lower dementia risk. Nutritional and physical variables such as MNAb_tot and waist circumference also show significant effects, with MNAb_tot negatively related and waist positively related to dementia status. Some variables like body height, body weight, and hyperlipidaemia were not statistically significant, implying limited impact in this model. Overall, the results emphasize the importance of demographic, nutritional, and anthropometric factors in predicting dementia risk.

The goodness-of-fit test on the logistic model reveals the following results:

Measure	Value
Log-Likelihood (llh)	-265.84
Null Log-Likelihood (llhNull)	-599.90
Likelihood Ratio (G2)	668.12
McFadden's R^2	0.557
Cox and Snell's R^2 (r2ML)	0.401
Nagelkerke's R^2 (r2CU)	0.666

Table 5: Pseudo R-squared values for the logistic regression model

The pseudo R-squared values provide measures of model fit for the logistic regression. McFadden's R^2 value of 0.557 indicates a strong model fit, suggesting that the model explains about 55.7% of the variation in dementia classification compared to the null model. Cox and Snell's R-squared (0.401) and Nagelkerke's R-squared (0.666) further confirm a good explanatory power, with Nagelkerke's measure adjusted to a maximum of 1, indicating that the model

explains roughly 66.6% of the variance in the outcome. The substantial improvement from the null log-likelihood (-599.90) to the model log-likelihood (-265.84) further supports the model's effectiveness in predicting dementia status.

The evaluation of the logistic model can be done by analyzing the ROC curve and also checking for the AUC which should be close to 1.

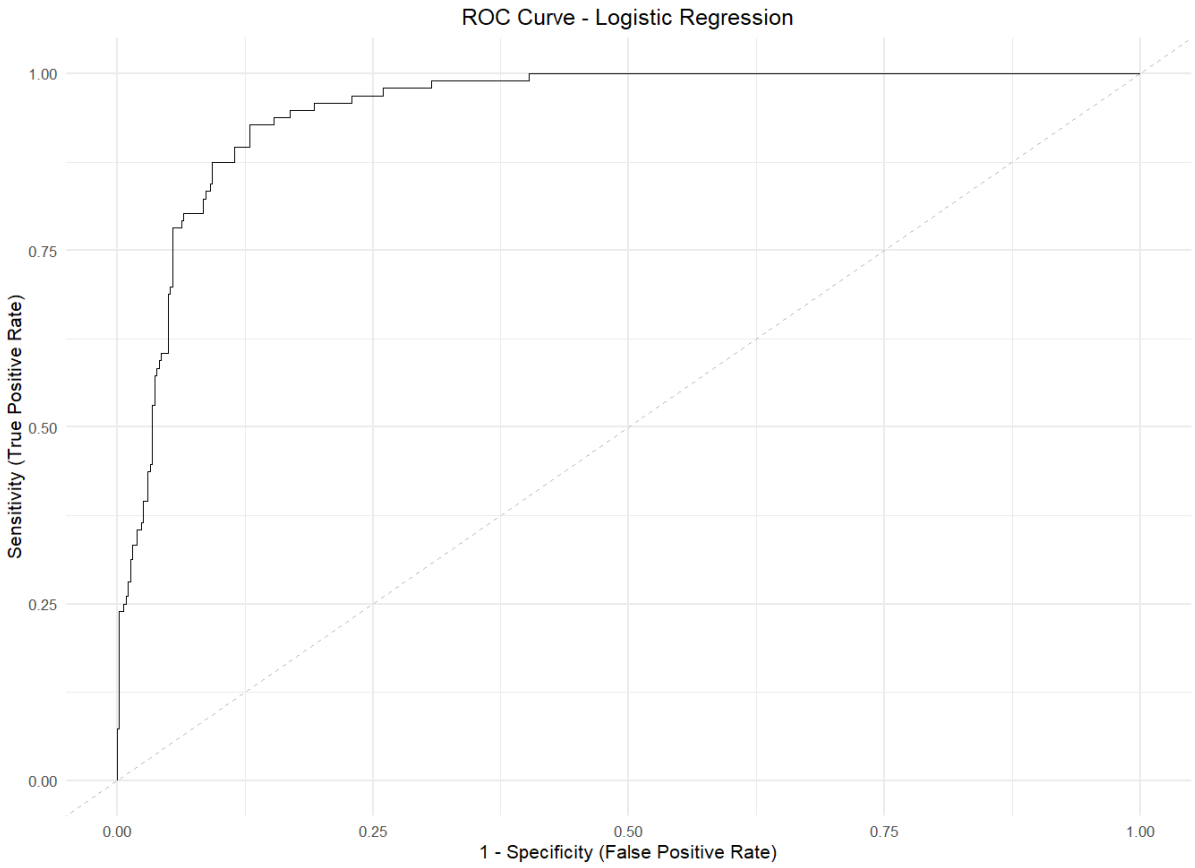

Figure 5: ROC curve for the logistic regression model predicting dementia status

The ROC (Receiver Operating Characteristic) curve illustrates the model's ability to correctly distinguish between individuals with and without dementia. By plotting the true positive rate (sensitivity) against the false positive rate (1-specificity) at various threshold settings, the curve shows how well the logistic regression model performs across different classification cutoffs.

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**AUC**
Logistic Regression	0.91	0.72	0.78	0.75	0.95

Table 6: Performance metrics for the Logistic Regression model predicting dementia status

The Logistic Regression model demonstrates strong overall performance in predicting dementia status. It achieves an accuracy of 91%, indicating that it correctly classifies the majority of cases. The precision of 72% shows that when the model predicts dementia, it is correct most of the time. The recall value of 78% reflects the model's ability to identify actual dementia cases

effectively. With an F1 score of 0.75, the model balances precision and recall well. Finally, the high AUC of 0.95 confirms excellent discriminative ability, meaning the model reliably differentiates between dementia and non-dementia individuals across different thresholds.

8.3 Conclusion

This study successfully developed a logistic regression model to predict dementia status using demographic, physical, and nutritional variables. The model demonstrated strong predictive performance, with high accuracy, precision, recall, and an impressive AUC, highlighting its ability to effectively discriminate between dementia and non-dementia cases. Key predictors such as age, education level, nutritional status, and anthropometric measures showed significant associations with dementia risk, underscoring the multifactorial nature of the disease. These findings emphasize the potential of integrating clinical and lifestyle factors for early dementia detection, which could inform timely interventions and improve patient outcomes. Future work may explore advanced machine learning techniques and larger, more diverse datasets to enhance prediction accuracy and generalizability.

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Appendix

Source Code in R

Listing 1: R Code for Dementia Prediction

```
# Regression Project
# Dementia Risk Analysis
# Author: Kabita Joshi

# Load necessary libraries
install.packages("DescTools")
install.packages("pscl")

library(pscl)
library(DescTools)
library(tidyverse)
library(caret)
library(randomForest)
library(pROC)
library(ggplot2)
library(readxl)
library(RColorBrewer)
library(reshape2)

# Set random seed
set.seed(12)


# 1. Load Dataset
data <- read_excel("dementia.xlsx")
head(data)

# View structure
str(data)
colnames(data)

# 2. Data Preprocessing
colSums(is.na(data))  # Check missing values
data_clean <- na.omit(data)  # Remove rows with missing values

# Convert categorical variables
data_clean$Education_ID <- as.factor(data_clean$Education_ID)
data_clean$Mobility <- as.factor(data_clean$Mobility)
data_clean$Hyperlipidaemia <- as.factor(data_clean$Hyperlipidaemia)
data_clean$MMSE_class <- as.factor(data_clean$MMSE_class)
```

```r
# 3. Exploratory Data Analysis (EDA)
summary(data_clean)

# --- Enhanced Correlation Matrix using ggplot2 ---
numeric_vars <- data_clean %>% select(where(is.numeric))
corr_matrix <- cor(numeric_vars)
melted_corr <- melt(corr_matrix)

ggplot(melted_corr, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "red", high = "blue", mid = "white", midpoint
     = 0,
                       limit = c(-1, 1), space = "Lab", name = "Correlation
                          ") +
  theme_minimal(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
  labs(title = "Correlation␣Matrix", x = "", y = "")

# MMSE_class distribution
ggplot(data_clean, aes(x = MMSE_class)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution␣of␣MMSE␣Class", x = "MMSE␣Class", y = "Count")
      +
  theme_minimal(base_size = 14)

# Age distribution
ggplot(data_clean, aes(x = Age)) +
  geom_histogram(binwidth = 2, fill = "lightgreen", color = "black") +
  labs(title = "Age␣Distribution", x = "Age", y = "Count") +
  theme_minimal(base_size = 14)

# 4. Data Splitting
trainIndex <- createDataPartition(data_clean$MMSE_class, p = 0.7, list =
   FALSE)
train_data <- data_clean[trainIndex, ]
test_data <- data_clean[-trainIndex, ]



## Logistic Regression
log_model <- glm(MMSE_class ~ ., data = train_data, family = binomial)
log_pred_prob <- predict(log_model, test_data, type = "response")
log_pred <- ifelse(log_pred_prob > 0.5, "1", "0") %>% as.factor()

#Significance Testing and Goodness-of-Fit for Logistic Regression
# --------------------------------------
```

```r
# 1. Likelihood Ratio Test (overall model test)
null_model <- glm(MMSE_class ~ 1, data = train_data, family = binomial)
lrt <- anova(null_model, log_model, test = "Chisq")
print("Likelihood Ratio Test:")
print(lrt)

# 2. Wald Test (coefficients significance)
summary_log_model <- summary(log_model)
print("Wald Test (Coefficient Significance):")
print(summary_log_model)

# 3. Goodness-of-Fit: Pseudo R   (McFadden) and Nagelkerke R

pseudo_r2 <- pR2(log_model)
cat("\nPseudo R-squared values:\n")
print(pseudo_r2)


# Evaluation for Logistic Regression
log_cm <- confusionMatrix(log_pred, test_data$MMSE_class, positive = "1")
log_precision <- posPredValue(log_pred, test_data$MMSE_class, positive = "1
    ")
log_recall <- sensitivity(log_pred, test_data$MMSE_class, positive = "1")
log_f1 <- 2 * log_precision * log_recall / (log_precision + log_recall)
log_roc <- roc(test_data$MMSE_class, log_pred_prob)



# 6. ROC Curves (Corrected: x = 1 - Specificity)

## ROC for Logistic Regression
ggroc(log_roc, legacy.axes = TRUE) +
  geom_abline(linetype = "dashed", color = "gray") +
  labs(title = "ROC Curve - Logistic Regression",
       x = "1 - Specificity (False Positive Rate)",
       y = "Sensitivity (True Positive Rate)") +
  xlim(0, 1) + ylim(0, 1) +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(hjust = 0.5))




# 7. Performance Summary Table
results <- data.frame(
  Model = c("Logistic Regression"),
  Accuracy = c(log_cm$overall["Accuracy"]),
```

```
  Precision = c(log_precision),
  Recall = c(log_recall),
  F1_Score = c(log_f1),
  AUC = c(auc(log_roc))
)

print(results)
```

Code Dependencies

- R version 4.3.1 (2023-06-16)
- `ggplot2` 3.4.2 for visualization
- `lmtest` 0.9-40 for diagnostic tests
- `jtools` 2.2.2 for model summaries